LEO-2100Ш

ELECTRONIC COLPOSCOPE

User's Manual

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Preface

Before using the apparatus, user should read this text carefully. You will be told how to take care of your operational process, which could not lead to abnormal operation and harmful accident. If the accident happened, which can absolutely be avoided, our company will not be in charge of the responsibility. And not for free service.

Advice:

- The electronic colposcope should be sold by professional salesmen.
- The electronic coposcope should be used by eligibility user.

Notice for the user:

- Thank you for your purchasing of this Electronic Colposcope, and we will try our best to offer your good service. In order to use the system correctly and safety, you should read and comprehend all the items in this manual before operation, and review regularly.
- You should pay more attention to the contents with “WARNING”, “NOTES”, “PROMT”
- The user should operate the system according to the instructions in the manual and the specifications of the authorized agent. The user may be responsible for any harm or damage caused by the disoperation of the user.
- The following signs are specified special meaning in the manual.

Safety sort and sign:

- All sorts of signs for potential danger are listed below according to medical electrical safety device signs founded by IEC：

Anti-electronic shock sort: type I device。
Anti-electronic shock grade：type B。

The “I” and “O” denote “ON” and “OFF” of the power respectively

The sign represent attention for safety：

1、 Use conformity power, especially notes the parameters of power, voltage and frequency.
2、 Use Grounding socket.

The sign represents equal electricity potential circuit for conductor connection.

◆ The following signs used in the internal of the facilities：

Indicate connection point for safety grounding.
According to device safety requirement, conductor part of Grade I device should not be connected to safety groundling.

Prompt：

This symbol means information of installation, operation and maintenance. The information itself doesn't have any danger, but it is important to correct operation.

Attention： This symbol means potential danger. Improper environment or incorrect operation will result:

- Minor damage
- Property loss

Warning： This symbol means client to keep the system working under normal scope. Incorrect operation will result：

- Serious hurt to people
- Fatal property loss

General safety items：

◆ Safety standard GB9706.1 are used in the system. The safety sort of the system is Grade I and Type B. Please obey the following rules for the safety of the user:

1、 The system should be used by eligible user.

2、 The patient need not have contact with the system at any time.

3、 Don't change internal parameters of the system. If you insist on changing, please contact with our company or our agent.

4、 The system has been adjusted to the best. The user need not adjust any undescribed switch.

5、 If there is any failure in the system, please turn off the system and contact with our company or authorized agent.

6、 If other electronic or mechanical devices should be connected to the system, please contact our company and authorized agent.

7、 Avoid the below environment：

—— temperature: below negative 5℃, above40℃
(suggested temperature：5℃～40℃)
—— atmosphere pressure: below 86kPa, above106kPa
—— poisonous gas
—— relative humidity, more than80%
—— water vapor
—— splashing environment
—— dust environment
—— petrol gas
—— salt fog
—— explosive gas
—— shock and vibration
—— slant angle of the system over ten degree
—— unsuitable or unstable power
—— vibration
—— direct sunshine
—— poor ventilation

8、 Environmental Requirement of Transportation and storage：

Environment temperature: -40℃～+70℃；
Relative humidity: 10%～100%, condensation；
Atmospheric pressure: 500hPa～1060hPa；

9、 Environmental Requirement of Operation：

Environment temperature: +5℃～40℃；
Relative humidity: ≤80%
Atmospheric pressure: 700hPa～1060hPa；
Power: AC220V±22V，50Hz±1Hz
Input power：≤300VA

Warning：

■The apparatus should be kept horizontal. There should be 1 or 2 minutes between 2 times of turning on the power.

■After one patient is checked, the source should be turned off.

● Indication and contraindication

Indication： It can be used in the examining of vulva, vagina and uterine neck disease

Attention：

If there is any abnormal situation in the working process, please

turn off the power.

Notice for turning on the system:

Preparation for turning on the system

1、 Check the power. Install AC regulator if the power can't meet the required conditions for the system.

2、 Check all the interfaces.

3、 Check the connection of grounding cable of main unit.

4、 Make sure the switch of the main unit is in "O" position (OFF). Connect the power plug to the power socket.

Procedure of operations

1、 Turn on the power after carefully checking of the power and all the interfaces.

2、 Operate after starting the system

Procedures of turning off the system

1、 The source camera should be turned off.

2、 The system should be turned off.

3、 Unplug power

4、 Remove the electric plug from an outlet.

Rubbish processing:

- The rubbish in the system operation procedure such as the packages and all the rubbish should be processed according to the regulations of the environmental protection agency.

Chapter 1 Overview

LEO-2100I Electronic Colposcope system is high-tech equipment with the technology of high resolution and clarity, which is combined by light, electricity and engine. The device is used for clinical detection of pathological changes in vulva, vagina and cervix.

It adopts hi-quality CCD digital camera system, annular surgical light source and enlarged image, which can discriminate extremely small focus of a disease, and analyze, save and print through the powerful software system.

The apparatus has implemented clinical test, and it is secue and effective.

Chapter2 Features and Proper Index

2.1 Features:

- High quality imported digital camera system
- Hi-frequency automatic focusing and adjustment:

When the camera can be adjusted automatically according to the distance to objects and intensity of the environmental light. And meanwhile, it can offer manipulating assistant adjust function.

- Convenient bracket offers an advanced environment for the examination and operation.
- Computer-assisted demonstration, collection, printing and management; offers an advanced environment for modern clinical diagnosis and long distant consultation

2.2 Proper Index:

Image interface: Video

Minimum illumination: ≤0.01 Lux

Camera resolution: ≥ 500 TVL

Focus: 10 mm～400 mm

Optical amplify: ≥ 22 times

Electronic amplify: ≥ 10 times

Pixels: ≥ 830000 pixels

Scope of illumination: ≥ 1400 Lux

System resolution: ≥ 500 TVL

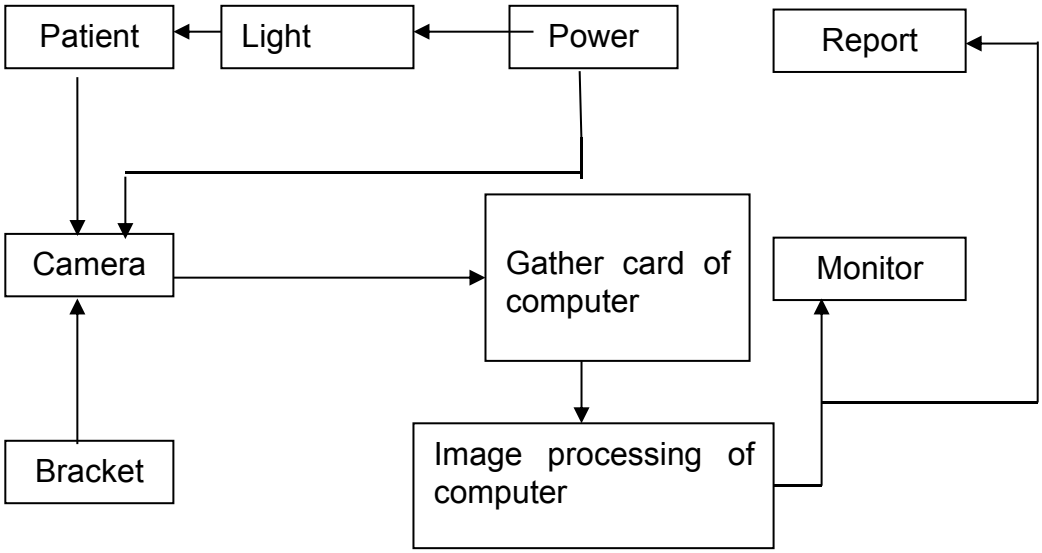
Operation humidity: ≤ 80%

Operation temperature: 5℃ ∽ 40℃

Computer configuration:

CPU: ≥ 2.0 G Memory: ≥ 2GB
Hard disk: ≥ 320G

Chapter 3 Composition and Working Principle



Schematic Diagram

With the light source illustration, the camera can transfer the collected pictures to main unit for process and analysis. Through the high resolution display, legible tissue images can be shown for doctor's discern. Images and reports are available for storage and print.

Chapter 4 Operation Instruction

4.1 Operations for buttons of image capture system:

4.1.1 While examination of nidus, press the button *zoom in* or *zoom out* to enlarge or reduce images.
4.1.2 Press *filtering+* and *filtering-* to adjust filtering process. Press *white-balance* to revert to original images.
4.1.3 Press *freezing*, operators can collect the picture displayed in monitor. Continuous collection is available. Footswitch has the same function with *freezing* button for the convenience of operators.
4.1.4 Press *focus+* and *focus-* to adjust focus.

4.2 Software Operations

How to enter operation interface

After computer startup, double click the colposcope icons at desk, it will enter our home page of the software. Right click the mouse, and you'll enter operation interface.

Operation interface of the software

4.3 Examination:

4.3.1 When pressing *New case history*, the equipment will clear all the records of last patient automatically. Then operator can input the new patient's information or press *Detail* to input more information. Meanwhile the apparatus will automatically put through the video port to receive signals for image display.

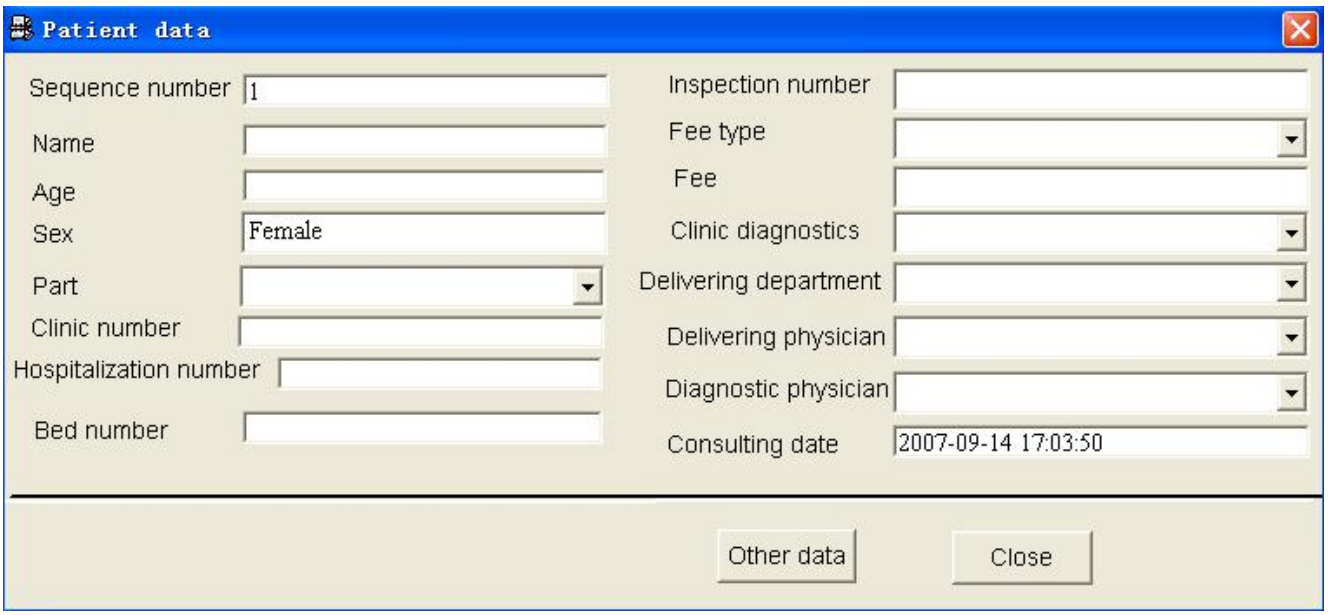
Patient's information is inputed in the widows, which include:

A. Number: the number is displayed by system automatically, unnecesary for mnual input.
B. Name: client name.

C. Gender: Defauted as female..

D. Age: client age.

The detail of the patient:



Input mode of the patient's detail:

Press 【Ctrl】 + 【spaces】: change into Chinese input method, input the information of the patient

Press 【Ctrl】 +【Shift】change the input mode

Press 【Snatch】, the image can be saved in the dynamic gallery. Right click in the dynamic gallery, each picture can be removed to display area. If operator chooses【show 4 in the screen】on any picture, the four pictures in sequence in the gallery will be displayed.

4.3.2 Remove the scroll bar, all of pictures in dynamic library can be browsed. If operator wants to view any picture in the dynamic library, he can right click on it and choose any function for process.

4.3.3 Department Setting: at bottom of this interface, there are【Original image】,【Image reverse】,【Left-right image】,【Left-right reverse】,【Up-down image】,【Up-down reverse】 and【Dynamic pseudo-color】 options for operator's convenient operation.【Original picture】 is to help operator recuperate the picture to avoid misjudgement after observation.

4.3.4 Click the brown frame at bottom of operation interface for complementary color.

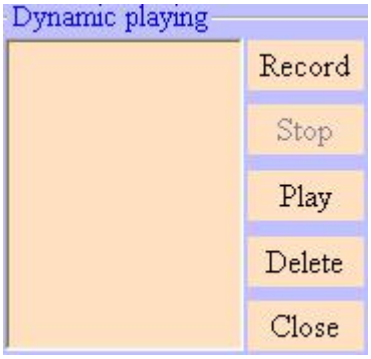
4.3.5 Press 【Save report】, the window similar as below will be ejected. Operator can choose yes or not for storage.

4.3.6 Magnification and storage the collecting image:

If doctor wants further view, right click on the image he chooses in dynamic gallery and select "Image enlarge". Right click again on the enlarged image, operator could choose primary, double, treble and fourfold magnifier rate and click on the local image area.

4.3.7 Record the image:

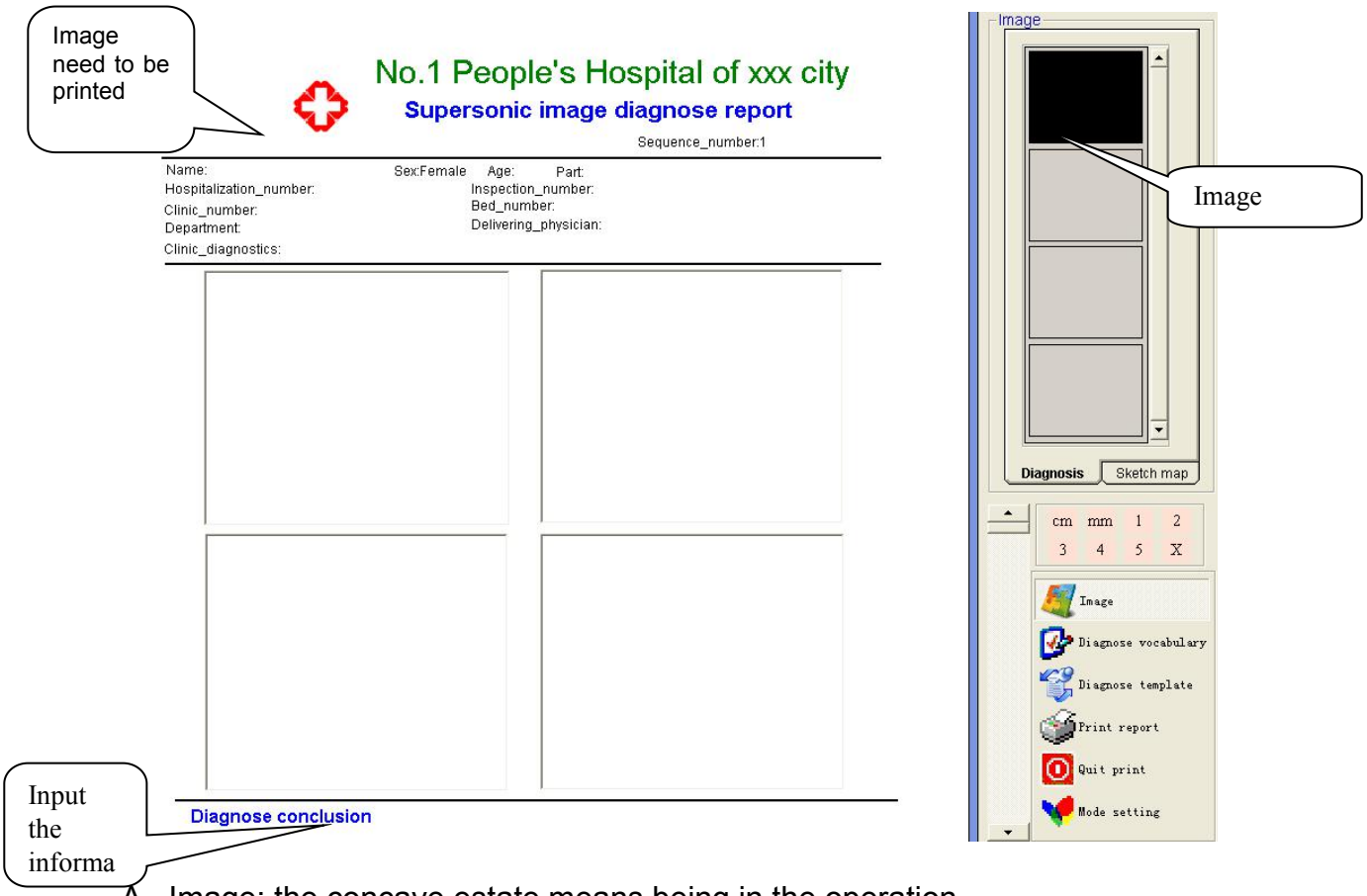
Click【Record image】, the screen will show the following window:



Click "Record", a big window appears over the top and shows the state of record, and a small one will show the content of the recording. Press【Stop】 to end the record. Left click the record you want to play and choose【Play】, doctor could view the records.

4.4 Print the report

Click 【Print report】to enter the printing interface(as below)



A. Image: the concave estate means being in the operation.

B. Diagnose vocabulary: click to select and add appropriate vocabularies.

C. Diagnose templete: there are common case history pictures, diagnosis conclusions and diagnostic templet.

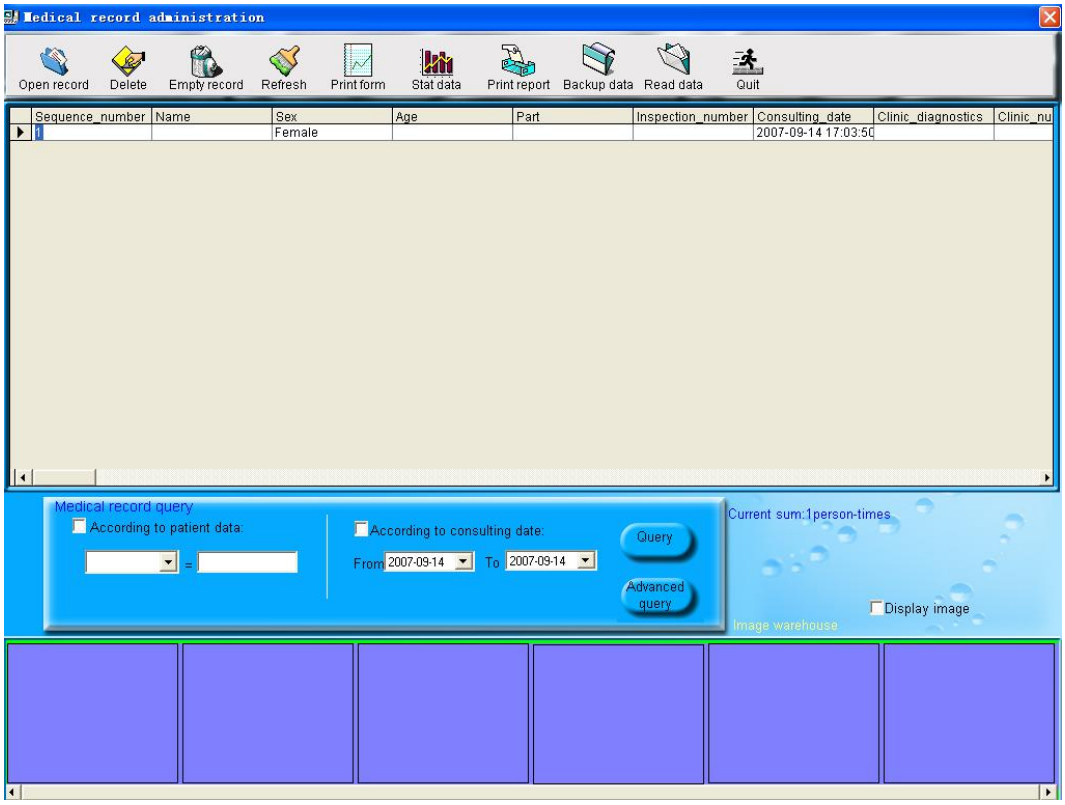
D. Print report: click *print report* to print the filled report.

E. Quit print setting: Back to the original interface

F. Mode setting: operators could modify as needed after entering modification interface and then save. Click *Save report* after back to original interface, system will save newly modified contents as well.

4.5 Medical record administration：

Click【Medical recordadministration】 shown as below, operators could browse, query, delete the profile saved previously.



4.5.1 Select *Sequenc number* and click *Open record* and *Display image* (interface down right corner), operator could review images saved in dynamic gallary previously.

4.5.2 Delete: delete the selected case history.

4.5.3 Empty records: delete all records in administration system. (Cautions for this function)

4.5.4 Refresh: if certain record items still exist after record deleted, please click *Refresh* to clear.

4.5.5 Print form: print the generated reports.

4.5.6 Stat. data: statistics of all medical records according to doctor requirements and generate 2D bar chart.

4.5.7 Backup data: system will inevitably collapse after long time use.Operator shall backup medical recores regularly to keep records complete. Defaulted backup directory is D:\.

4.5.8 Read data: read the backup records after system collapse.

4.5.9 Medical record query: operator could choose to select *according to patient data* or *according to consulting date* and click *Query*.

4.5.10 Exit: back to original interface

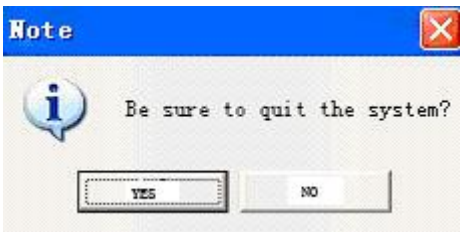
Promotion：

The operation of delete and empty record is physical operation.Deleted document can not be restored. Please be careful to using this function

Attention：

Please do not change parameters of Video setting and Department setting, which will relate to the normal operation of apparatus.

Press 【Exit】, the dialog box will be shown in the screen.



Choose “Yes” to close the running program.

Choose “No” to continue the program.

Attention:

Dear client, there may be some differences between the machine and information which we offer you in the manual since for the upgradation of software. LEO have the right to modify in any time, so we may not to tell all the customers. If you have any question, do connect with us in any time.

Prompt:

We guarantee the main unit one year from delivery date. Services in guarantee time are free of charge. We offer a lifetime service.

The following malfunction is not in free guarantee:

★ The malfunction caused by dismounting and reequip without authorization.

★ The malfunction caused by drop, knock in transport.

★ The malfunction caused by inappropriate operation.

★ The malfunction caused by maintainence without permitting.

★ The artificial damage (include computer virus, misoperation and damage in the transport)

★ The product goes beyond the guarantee time.

★ Please backup the important data and datum in time.

★ please get in touch with us or our distributor directly when it is necessary.

★ Our company promise to supply the necessary circuit diagram, the list of the parts and so on, if it is permitted.

★ TEL FOR SERVICE: 86-516-69695110
★ TEL：86-516-69695099/98/97/96
★ FAX：86-516-69695177

Chapter 5 Troubleshootings

5.1 No image in the monitor：

5.1.1 Possibility I: Power is off or the power line is loose.
Process: Turn on the power, or insert the power line tightly.

5.1.2 Possibility II: The display signal line is loose
Process: Insert the line tightly.

5.1.3 Possibility III: The brightness or the contrast is low.
Process: Regulate the brightness or contrast to make it brighter.

5.1.4 Possibility IV: The power line or the signal line of the camera is loose.
Process: Insert the line tightly.

5.1.5 Possibility V: cover of the camera is not taken off
Process: Take off the cover.

5.2 Image floating or can’t be seen clearly

5.2.1 Possibility I: The focus of the camera is no suitable
Process: Adjust the focus

5.2.2 Possibility II: Dynamic check is still on while freezed.
Process: Correct the operating method.

5.3 Light source does not work

5.3.1 Possibility I: Bad connect
Process: Tighten the light resource

5.3.2 Possibility II: The switch of the light resource is closed.
Process: Turn on the light resource.

5.4 The indicator light does not work

5.4.1 Possibility I: The power supply is abnormal.
Process: Check if the power of the socket is normal.

5.4.2 Possibility II: The fuse is damaged
Process: Exchange the fuse, which is behind the board of the main unit.

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